



Power modules

vendor who had been doing this manually was desperate to have the solution working.

On investigation, it was found that the boards had not been soldered properly, leading to the components changing their characteristics (the design also had flaws with low-accuracy parts). Above all, there was no provision for calibration, which led to the wrong readings. After 6-8 weeks of struggle, the battery manufacturer called

off the trials and even today continues with manual measurements.

The second example shows how a wrong assembly can fail the product for compliance standards. An OEM AC-DC power supply was selected for integration into our product. After integrating the PSU into our design, we took the product for compliance testing. The product failed in the conducted emission test consistently.

We finally called the PSU OEM as our experience told us that conducted emission failures are typically due to power supply related circuits. The PSU OEM opened and found that the 220V AC lines were connected as two individual wires. They just twisted the phase and neutral wires and the problem disappeared. Later we discovered that they had told our design team about this, but our design team had failed to inform the manufacturing team.

What needs to be factored in

The key to successful product development is to have a holistic view of the overall process. It is important that the EMS partner is selected and brought on board early on in the development process to ensure that the design is manufacturable. The processes followed by big EMS are extensive and they invest a lot in these processes. But many MSMEs and startups work with Tier 2 or Tier 3 EMS vendors, who may have limited capacity and are typically just assembly houses.

Large EMS vendors typically do what is known as fulfilment. When we choose this mode, EMS partner does everything from component pro-

curement to final delivery of finished product to the customer. Tier 2 and Tier 3 vendors do not offer this, so when you work with them you have to plan and implement the design covering the aspects needed for the manufacturing.

This approach helps but may not solve the issue completely. For example, an ICT tester will cost \$200,000 to \$300,000, and the 5D XRAY equipment for BGA soldering inspection costs close to \$300,000. Designers have to understand the machines that their manufacturing partners have and implement the design accordingly. This means the design teams have to know the manufacturing process well and plan to complement the skill that is not available in the EMS partner's line.

Points to keep in mind

Electronics product manufacturing is a vast topic, which is also dependent on the design. Every electronic product is unique, and the manufacturing needs to align with that. While giving an exhaustive list of points to note is out of the scope of this article, here are some key points that are important.

1. Evaluate the EMS partner based on your product's need and match them. For example, if your product uses lots of magnetics, an EMS vendor who can make them inhouse will be preferable to an EMS vendor who buys it from other vendors.

2. If your product is medical or avionics related, the EMS partner should have relevant certifications and also IT and other infrastructure (for example, medical devices manufacturing needs sterilisation facility and Clean Rooms).

3. Empanel EMS partners who use manufacturing machinery from established manufacturers.

4. When transitioning the design to the manufacturing line, give robust training for the product and ensure everyone involved in the product's manufacturing is trained.

5. Continuously monitor the product's manufacturing performance like test process yield, assembly yield, obsolescence, and custom parts performance. **EFY**



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too to test the product and also help in calibration if the product has any measurement function.

Another key aspect that designers leave unaddressed is the product packing for shipment. A well-designed product can be damaged in transit if the product shipment package is not designed properly. It is seen that the product shipment packaging is never addressed and it gets attention only when customers report damaged product.

Startup entrepreneurs and MSME owners should address the issue of manufacturing as an integral part of the design process and design the product in consultation with the EMS partners. This means identification of EMS partner is crucial and needs to be done early in the product development phase.

Some examples

To exemplify the importance of manufacturing here are two real-life examples. The first is a test & measuring equipment designed for a battery manufacturing line to measure cell voltages during the battery manufacturing process. This equipment records the readings through wireless connections into the cloud (both Wi-Fi and Bluetooth). Since each batch has 100 cells, each cell has to be monitored individually.

The company that supplied the equipment had developed a coin-size unit connected to every cell. They had supplied the first batch of 200 units for trial. While the software (both cloud and firmware) was working properly, the readings were inconsistent. The issues ranged from intermittent working to wrong readings. The battery